

Day 2: Variables (Like Storage Boxes)

JavaScript Variables - Notes

☐ Real-Life Example: Lunch Box ☐

☐ "Clg ki vache mundhu Lunch box lo different items pack chestav...
JavaScript lo kuda different data store cheyyali ante variables use chestam!"

Variables **store different types of data** like numbers, text, and true/false values.

☐ C vs JavaScript - How Variables Work?

☐ "C lo `int`, `float` lantivi use chestam... Kani JavaScript lo `var`, `let`, `const` tho variables declare chestam!"

C Language (Data type fix)

```
c
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int x = 10; // Number only
float y = 3.14; // Decimal number
char name = 'A'; // Single character
```

JavaScript (Dynamic - No fixed type)

```
js
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let x = 10; // Number
let y = "Vamsi"; // Text (String)
let z = true; // Boolean (true/false)
```

C lo data type fix → JavaScript lo auto adjust! ☐

☐ Types of Variables in JavaScript

☐ "Variable ante oka container laga values store cheyyadaniki use chestam!
Suppose mana degara 3 boxes unayi—`var`, `let`, `const`..."

🔗 Box Examples:

Variable Type	Can Change Value?	Scope Control?	Recommended?
var	☐ Yes	☐ No	☐ Avoid Using
let	☐ Yes	☐ Yes (Block Scope)	☐ Use for normal values
const	☐ No	☐ Yes (Block Scope)	☐ Use for fixed values

☐ Simple Explanation:

- **var box** – Open chesthe values replace avuthayi, kani use cheyyadam recommend cheyyaru!
- **let box** – Open chesthi values replace cheyyachu, but scope control untadi!
- **const box** – Once value pedithe, forever fixed! Change cheyyalem!

☐ Example Code: var, let, const

```
js
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// Box - var (Can change, but avoid)
var box = "car";
console.log(box); // car
box = "Bike"; // 🔗 Value changed
console.log(box); // Bike

// Box - let (Can change, safer)
let box1 = "car";
console.log(box1); // car
box1 = "Bike"; // 🔗 Value changed
console.log(box1); // Bike

// Box - const (Once stored, can't change)
const box2 = "Bike";
console.log(box2); // Bike
box2 = "car"; // 🔗 ERROR! Can't change
```

☐ How to Check a Variable's Type?

□ "C lo manam variable declare chestappude data type fix untadi (`int x = 10;` ante integer ani telustadi).

Kani JavaScript lo variables dynamic ga untayi—data type runtime lo decide avuthadi!"

So, data type teliyali ante `typeof variableName` use chestam!

Example:

js

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```
var x = 10;
console.log(typeof x); // "number"

var y = "Vamsi";
console.log(typeof y); // "string"

var z = true;
console.log(typeof z); // "boolean"
```

□ C lo compile time lo data type fix... JavaScript lo runtime lo `typeof` tho check cheyyali!

□ □

□ Key Takeaways

- ✓ □ JavaScript variables store different types of data dynamically.
- ✓ □ Use `let` for normal values & `const` for fixed values.
- ✓ □ Avoid using `var` (it has scope & hoisting issues).
- ✓ □ Check the data type using `typeof variableName`.

Every great coder starts with the basics! Master variables today, and tomorrow, you'll be writing logic like a pro!